

Online Resource 1: Primary-study evidence matrix

Companion to From Checked Artefacts to System Assurance: A Critical Review of Trust Boundaries in Generative AI and Formal Methods. Prepared 23 September 2026. This is a purposive comparison of 20 studies or study families, not an exhaustive screen or a frequency estimate. Eighteen entries came from the retained full-text comparison register; Astrogator and Alloy tests were added by targeted title/topic search. ACCESS and the continuous-assurance CLARISSA study are contextual, non-generative or mixed precedents. A publication and its preprint are counted once. Locations point to the linked version; claims are attributed to the papers unless this review independently checked them. The authors should confirm each interpretation against the final submission versions.

The four appraisal columns ask what evidence a broader deployed-system claim would need: S specification adequacy, M semantic correspondence between representations, O mediation of live observations and effects, and C composition of results under environmental assumptions. E means the study explicitly supplies evidence for that link within a stated scope; P means a bounded test, review or partial relation is supplied; R means a broader reuse claim relies on this link as an assumption or external input; – means the link is outside the study’s declared objective. These are this review’s claim-specific appraisals, not ratings of paper quality. A dash is not a failure. Even E is not a universal guarantee.

Role codes: A generate/check; B generate/prove; C model-based or empirical assessment; D formalism serving the model; E runtime enforcement; F assurance-oriented. Codes may overlap and do not change the one-label arXiv map.

Study (role); source locator	Accepted object and authority	Supported result and residual obligation	S	M	O	C
Clover (A/C), §§3.1–3.2, 4, 6, App. 0.A.1	Code, annotations and docstrings; Dafny and reconstruction consistency	Contract conformance and selected consistency relations; jointly generated artefacts can still miss user intent; unit tests are future work	R	P	–	–
FVEval (A/C), §§3.2–3.4	Generated hardware assertions; Jasper equivalence/implication checks against RTL/reference	Formal benchmark verdict; adequacy of reference assertions and environment remains an input	R	E	–	–
Lahiri et al. (A), §§I-C, II-A–C	Candidate Dafny specifications; Dafny plus reference tests/mutations	Detects some weak or wrong specifications in evaluated examples; finite tests/mutations do not certify full intent	P	–	–	–
Astrogator (A), §§3–4, 5.2–5.5, 6.3	Generated Ansible and a formal query (hand-written in evaluation; user review proposed); symbolic verifier	Query satisfaction within supported Ansible/query semantics; user approval of underspecified choices is proposed, not evaluated end to end	P	E	–	–
Cunha–Macedo Alloy tests (A/C), §§3.2–3.4, 4.4–4.6	Generated positive/negative examples; Alloy and hidden oracle	Empirical counterexamples to some incorrect small-domain specifications; scenario coverage and language ambiguity remain	P	–	–	–
TR2MTL (A), §§IV–V	Traffic rules translated to MTL; parsing/reference-formula evaluation	Translation quality for studied rules; interpretation, perception and map assumptions remain	P	–	–	–
Herald (B/A), §4.1.2, App. D, Table 4	Informal-to-Lean statements; Lean validation, back-translation/NLI and separate human review	Formal acceptability and reviewed fidelity on a selected validation-passed set; compilation alone does not certify meaning. Of 151 reviewed passes, 101 were judged correct, 24 minor-error and 26 major-error; this is not a random quality-rate sample	P	–	–	–
DeepSeek-Prover-V2 (B), §§2.1–2.3, 3.2	Lean subgoals and proofs; Lean kernel	Derivability for fixed formal targets and permitted premises; revised results exclude misformalised items and correct an appLy? interface bug that sometimes concealed sorry declarations	R	–	–	–
SynCode (D), §§4.3–4.5; readable preprint	Grammar-constrained decoding; CFG/DFA mask	Valid-prefix extension under grammar/tokenisation conditions; not completion or program semantics	–	R	–	–
SafePlan (D/C), §§III–V	Robot plans/code; logic-structured model judgement	Empirical rejection and task performance; model judgement is not an independent formal acceptance verdict	P	–	P	–
PLC test generation (C), §§III–VI	PLC tests/assertions; compile/run, coverage and human inspection	Evidence for exercised behaviours; oracle adequacy and coverage limit generalisation	P	–	–	–
ToolGate (E), §§3.2, 3.4, Apps. E/G	Tool call and symbolic state; runtime pre/postconditions	Accepted outcomes enter internal symbolic state; rejected postconditions do not undo external tool effects	R	–	P	–
VeriGuard (A/E), §§3.2–3.3	Generated policy; offline Nagini/Viper proof and online action guard	Filtering under verified policy and correctly populated arguments; LLM extraction from live data remains a live mapping obligation	R	P	P	–
AESOP (E), §§III–IV, Thm. 1, Apps. E/H	LLM recovery advice; detector and model-predictive control	Conditional preservation of feasible recovery options; detector, dynamics and latency assumptions matter	R	–	P	P
Trusta (F/A), published version; preprint §3.2	Formalised assurance relations; Prolog/Z3/MONA	Logical analysis of encoded constraints; extraction fidelity and leaf-evidence truth remain external	R	–	–	P
ACCESS (F; contextual), §§3.4–3.7, 5.6–5.7	Engineering evidence and dynamic argument; runtime evaluation service	Status of modelled assurance nodes; service is non-invasive and does not interlock each action	R	–	P	P
CAF (F), PDF pp. 3–6	Formal safety assertions/dependency graph; proposed composition rules	Evidence propagation and deployment gate; aggregation needs property- and operator-specific justification	R	–	–	P
CLARISSA semantic analysis (F/A; mixed), §§5.1–5.4	Extracted object/property/environment relations; s(CASP)	Logic checks on encoded semantics; LLM extraction, evidence truth and system behaviour are distinct	R	P	–	P
CLARISSA continuous assurance (F; contextual), PDF pp. 9–11	Modelled evidence dependencies; ETB2 change detection/re-evaluation	Incremental rechecking of known dependencies; unknown dependencies and per-action authorisation lie outside the mechanism	R	–	–	P
Forge v2 (A; author-related), §§3–4, 5.5–5.7	Generated Java transformed to formal models; Dafny/Z3, FDR4 and Isabelle	Separate checks of encoded obligations; general source/model preservation remains unproved. All five review authors co-authored this study	R	P	–	P

Cross-study inferences and limits

Specification validation is an active research area: Astrogator proposes query review, Lahiri et al. and the Alloy tests probe candidate specifications, and Herald reviews a selected set of formal translations. None makes all user intent machine-checkable. Offline/online coupling is also real: VeriGuard checks a policy before deploying an interceptor, AESOP combines slow advice with fast recovery control, and ToolGate constrains symbolic state changes. Their live data mapping, latency and external-effect assumptions differ. ACCESS, CAF and both CLARISSA studies already connect or maintain assurance evidence; the question for a given system is the truth and completeness of the represented links. These observations follow from purposive cases and cannot establish how often any link is addressed across the field.

Reproducibility and provenance index

Online Resource 2 is the companion ZIP with the saved corpus_labelled.json, lit_search.py, label_corrections.json, check_counts.py, the corpus note, and the underlying comparison register. It preserves the 363-record arXiv denominator plus two tagged supplemental records; the counting script excludes the latter. lit_search.py records the fourteen historical query families and original classification prompt. The original retrieval timestamp, raw query responses, exclusion log, classifier version/settings and per-record rationales were not retained. The regular-expression repair and two targeted corrections do not restore that history or constitute independent human validation. The original targeted matrix is also in the ZIP as an audit record; the English extraction above is the reader-facing submission version.

The formal-artefact deposit remains a separate author decision. A curated deposit should include the relevant formal-artefacts/amr - * packages and cka-repair/, their manifests and English claim guides; historical Bridge_CKA.thy, Bridge_Progress.thy and Lean/Bridge.lean contain admitted or unfinished proof steps and must be segregated rather than presented as closed evidence. The 22 September record-maintenance pass recomputed some evidence checksums after requested postprocessing; those digests describe the maintained snapshot, not pre-execution files. No archive DOI, licence or independent author sign-off is asserted here.